

Disproof of conjecture 00000004124: the least size of a ccc, non-Knaster poset is never $\aleph_2$

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1 The conjecture and the clause refuted

Statement (English, verbatim). Definition: ccc, precaliber, and the Knaster property form the hierarchy of strengthening chain conditions of forcing. Conjecture: The least cardinality of a poset separating ccc from the Knaster property is $\aleph_2$, and the smallest counterexample can be constructed by refining a Suslin tree. (The Chinese text states the same: the least poset cardinality at which ccc and the Knaster property separate is $\aleph_2$, and the smallest counterexample is given by a refinement of a Suslin tree.)

Reading. The Knaster property implies ccc (Lemma 2), so a poset “separating ccc from the Knaster property” is a poset that is ccc but not Knaster. Let S be the class of cardinalities $|P|$ of such posets. The conjecture is a conjunction; its first conjunct says that the least element of S is $\aleph_2$. We refute that conjunct, which refutes the conjunction. We read “least cardinality is $\aleph_2$ ” in two ways and refute both:

(L1) $\aleph_2$ is the least element of S : some ccc non-Knaster poset has size $\aleph_2$ and every such poset has size at least $\aleph_2$;

(L2) $\inf S = \aleph_2$ (in Lean, `sInf`; the infimum of the empty set of cardinals is 0).

Each is refuted for three notions of “poset”: partial orders, preorders, and partial orders with a greatest element (the usual shape of a forcing notion). The second conjunct (a Suslin tree refinement) is not needed; see Section 4.

2 Definitions

Only the relation $\leq$ is used. For $p, q \in P$:

Definition 1. p, q are *compatible*, $\text{Comp}(p, q)$, if there is r with $r \leq p$ and $r \leq q$. An *antichain* is a set A such that any two distinct members of A are incompatible. A set B is *linked* if any two distinct members of B are compatible. P is *ccc* if every antichain in P is countable. P has the *Knaster property* (property K) if every uncountable $A \subseteq P$ has an uncountable linked subset $B \subseteq A$.

Sources (retrieved this session). Wikipedia, *Countable chain condition*: “a partially ordered set X is said to satisfy the countable chain condition, or to be ccc, if every strong antichain in X is countable.” Wikipedia, *Martin’s axiom*: “an antichain is a subset A of P such that any two distinct members of A are incompatible (two elements are said to be compatible if there exists a

common element below both of them in the partial order).” Y. Peng, *Fragments of Martin’s axiom*, arXiv:2510.21496: “a poset has property K_n (K for Knaster) if every uncountable subset of has an uncountable subset that is $\perp$ -linked where a subset is $\perp$ -linked if every $\perp$ -element subset of has a common lower bound” (math symbols lost in extraction; the Knaster property is the case $n = 2$, “Knaster’s original property K_2 ”). Peng also gives the equivalent form of ccc “every uncountable subset of contains two elements with a common lower bound”; Lean proves this equivalence (`ccc_iff`).

In Lean (`Conjecture4124`): `Compat`, `IsAntichainC`, `IsLinked`, `CCC`, `Knaster`, stated for any type with `LE`; antichains and linked sets use Mathlib’s `Set.Pairwise`, which only constrains distinct elements. Cardinalities are `Cardinal.mk`; $\aleph_1 = \text{aleph } 1$, $\aleph_2 = \text{aleph } 2$.

3 Proof

Lemma 2. *Knaster implies ccc. Every countable poset is Knaster. Hence every ccc, non-Knaster poset has $|P| \geq \aleph_1$.*

Proof. If A were an uncountable antichain, Knaster gives an uncountable linked $B \subseteq A$; B has two distinct elements, which are both compatible and incompatible. A countable P has no uncountable subsets, so it is Knaster vacuously. (Lean: `ccc_of_knaster`, `knaster_of_countable`, `aleph_one_le_of_not_knaster`.) $\square$

Lemma 3 (Key lemma, `exists_aleph_one`). *Let P be ccc and not Knaster, and $X \subseteq P$ with $|X| \leq \aleph_1$. There is Q with $X \subseteq Q \subseteq P$ and $|Q| = \aleph_1$ such that Q , with the order induced from P , is ccc and not Knaster.*

Proof. Since P is not Knaster, there is an uncountable $A \subseteq P$ with no uncountable linked subset. As $|A| \geq \aleph_1$, choose $W \subseteq A$ with $|W| = \aleph_1$; W is uncountable and also has no uncountable linked subset. Choose $f(p, q)$, a common lower bound of p, q whenever they are compatible (`lb`, `lb_spec`). Let $S_0 = W \cup X$, $S_{n+1} = S_n \cup f[S_n \times S_n]$ and $Q = \bigcup_n S_n$ (`stage`, `lbClosure`). By induction $|S_n| \leq \aleph_1$ (since $\aleph_1 + \aleph_1 \cdot \aleph_1 = \aleph_1$), so $|Q| \leq \aleph_0 \cdot \aleph_1 = \aleph_1$, and $|Q| \geq |W| = \aleph_1$ (`mk_stage_le`, `mk_lbClosure_le`). Q is closed under f (`lb_mem`), so for $p, q \in Q$: p, q are compatible in Q iff they are compatible in P (`compat_lbClosure_iff`). Hence an antichain of Q is an antichain of P , so it is countable: Q is ccc (`ccc_lbClosure`). And $W \subseteq Q$ is uncountable; an uncountable Q -linked subset of W would be P -linked, which is impossible, so Q is not Knaster (`not_knaster_of_subset`). $\square$

Theorem 4 (`conjecture4124_false`). *Fix a universe u , and let `sepCards`, `sepCardsPre`, `sepCardsTop` be the sets of cardinalities of ccc, non-Knaster partial orders, preorders, and partial orders with a greatest element (`OrderTop`) in `Type u`. For each of these sets S : $\neg \text{IsLeast}(S, \aleph_2)$ and $\inf S \neq \aleph_2$. Moreover, if $S \neq \emptyset$ then $\text{IsLeast}(S, \aleph_1)$.*

Proof. By Lemma 2, every element of S is $\geq \aleph_1$. If $S \neq \emptyset$, apply Lemma 3 with $X = \emptyset$ (or $X = \{\top\}$ for the third class, so that Q keeps the greatest element; Mathlib’s `Subtype.orderTop`); the induced order on Q is again a partial order or preorder, so $\aleph_1 \in S$ (`sepCards_spec` and its two variants). Then (`not_aleph_two`): if $\aleph_2$ were least, $\aleph_2 \leq \aleph_1$, false since $\aleph_1 < \aleph_2$; and $\inf S$ is $\aleph_1$ if $S \neq \emptyset$ and 0 if $S = \emptyset$, never $\aleph_2$. $\square$

So in every model of ZFC the least size of a ccc, non-Knaster poset is $\aleph_1$ if such a poset exists, and there is no least size otherwise. The conjecture’s value $\aleph_2$ never occurs. In particular the independence of the existence of such posets cannot rescue the claim.

4 The Suslin-tree clause

The second conjunct is not needed, since the first is false. For context only (not formalized): Wikipedia, *Suslin tree*: “a Suslin tree is a tree of height ω_1 such that every branch and every antichain is countable” and “Martin’s axiom $\text{MA}(\aleph_1)$ implies that there are no Suslin trees”. The standard argument (not formalized) is as follows. Order a Suslin tree T upside down, so that stronger conditions are higher nodes. Two nodes have a common extension iff they are comparable. So antichains in the forcing sense are tree antichains, which are countable, and T is ccc. Each level is an antichain, so $|T| = \aleph_1$. A linked set is a chain, and an uncountable chain would give an uncountable branch, so the uncountable set T has no uncountable linked subset and T is not Knaster. Thus the Suslin-tree example agrees with our result: it has size $\aleph_1$, not $\aleph_2$.

5 Lean formalization and axiom audit

File `lean/Conjecture4124/Basic.lean` (only import: `Mathlib`; 278 lines), Lean 4.33.1, `Mathlib v4.33.1`. Main theorem `Conjecture4124.conjecture4124_false`, with `exists_aleph_one`, `not_aleph_two`, `ccc_of_knaster`, `ccc_iff`. `lake build` succeeds, and `#print axioms` on `conjecture4124_false` and `exists_aleph_one` reports only `propext`, `Classical.choice`, `Quot.sound`. There is no `sorry` and no `native_decide`.